

# EE2703: Assignment 9

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## 1 Introduction

In this assignment, we continue our explorations on the DFT of a finite-length sequence with the FFT algorithm, using the `numpy.fft` module. We shall look at finding the DFTs of non-periodic functions, and problems associated with them, namely the *Gibbs Phenomenon* and how to overcome them, using *windowing*.

## 2 Spectrum of $\sin(\sqrt{2}t)$

Using the method we followed for obtaining the DFT of a periodic signal, we get the following spectrum for  $\sin(\sqrt{2}t)$  :

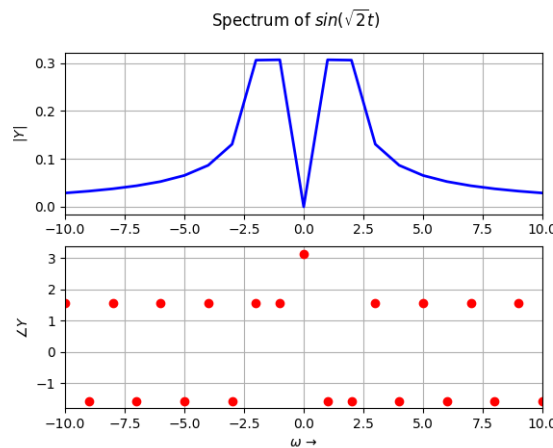


Figure 1: Spectrum of  $\sin(\sqrt{2}t)$

But, this is not what we expected! We expected two peaks, but that is not what we got. This is because, we aren't finding out the DFT of the required function,  $\sin(\sqrt{2}t)$  :

fig10-2.png

Figure 2:  $\sin(\sqrt{2}t)$

fig10-3.png

Figure 3: The function for which we are calculating DFT

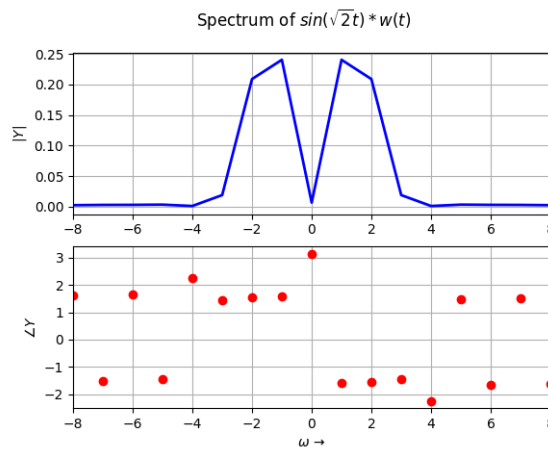
We can see that this is not what we want to calculate the DFT for. The discontinuities in the function has led to the very problematic *Gibb's Phenomenon*. So, we do not observe a sharp peak, but rather a flat one.

This problem can be fixed by **windowing**. Windowing is an operation in which we multiply the time-domain function with a suitable window function. In this assignment, we choose the *Hamming Window*, defined as:

$$W_N[n] = \begin{cases} 0.54 + 0.46 \cos(\frac{2\pi n}{N-1}), & |n| < N \\ 0, & \text{otherwise} \end{cases}$$

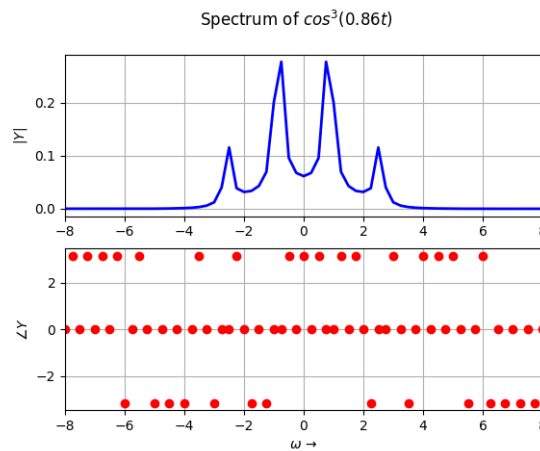
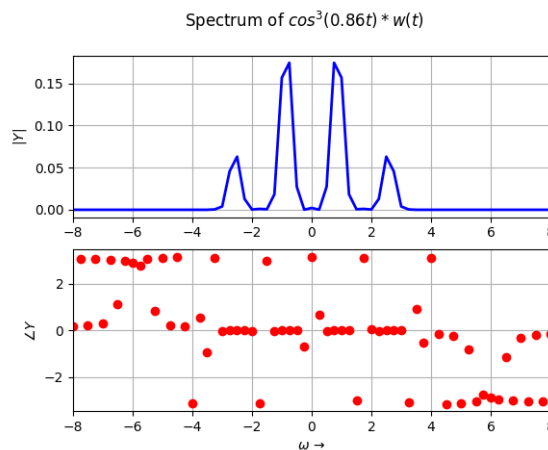
The result in time domain is that the magnitude of the jump is greatly reduced, thus minimizing the effect of Gibb's phenomenon in the frequency domain:

fig10-5.png

Figure 4: Windowed  $\sin(\sqrt{2}t)$ Figure 5: Spectrum of windowed  $\sin(\sqrt{2}t)$ 

### 3 Spectrum of $\cos^3(0.86t)$

We can see the effect of windowing even better for  $\cos^3(0.86t)$ . We get the following spectra before and after windowing:

Figure 6: Spectrum of  $\cos^3(0.86t)$  without windowingFigure 7: Spectrum of  $\cos^3(0.86t)$  after windowing

We can see narrower and sharper peaks at the frequencies that are present in the signal.

## 4 Parameter Estimation using DFT

We are given a 128-element vector with sampled values of the signal  $\cos(\omega_o t + \delta)$ , where  $\omega_o$  and  $\delta$  are to be estimated.

We cannot extract  $\omega_o$  from the DFT spectrum because the sampling rate is very low (only 128 points). The peaks will overlap and so we will not be able to extract their values.

I used an approach as illustrated below:

$$\omega_{o_{est}} = \frac{\sum_{\omega} |Y(j\omega)|^k \cdot \omega}{\sum_{\omega} |Y(j\omega)|^k}$$

$$y(t) = A \cos(\omega_o t) - B \sin(\omega_o t)$$

$$\delta_{est} = \arctan\left(-\frac{B}{A}\right)$$

In the above equations, I used a least squares approach to find A, B.

I varied the parameter  $k$  manually to find out which worked best for the given range of  $\omega_o$ . I used  $k = 1.7$ , for estimation in absence of noise, and  $k = 2.4$ , for estimation in presence of noise, to give the following results (I had set  $\omega_o = 1.35$  and  $\delta = \frac{\pi}{2}$  in both cases) :

| Noise         | $\omega_o$ | $\omega_{o_{est}}$ | $\delta$ | $\delta_{est}$ |
|---------------|------------|--------------------|----------|----------------|
| 0             | 1.35       | 1.453              | 1.571    | 1.571          |
| 0.1*rand(128) | 1.35       | 1.437              | 1.571    | 1.571          |

## 5 DFT of chirp $\cos(16t(1.5 + \frac{t}{2\pi}))$

A chirp is a signal in which the frequency increases or decreases with time<sup>1</sup>. We are going to analyse the chirp signal  $\cos(16t(1.5 + \frac{t}{2\pi}))$ . First, we shall plot the signal versus time, to get an idea of how the signal looks like:

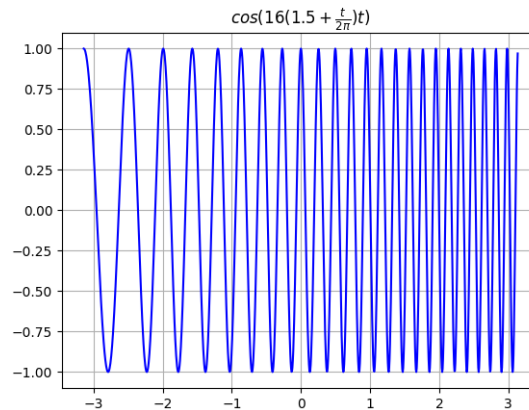


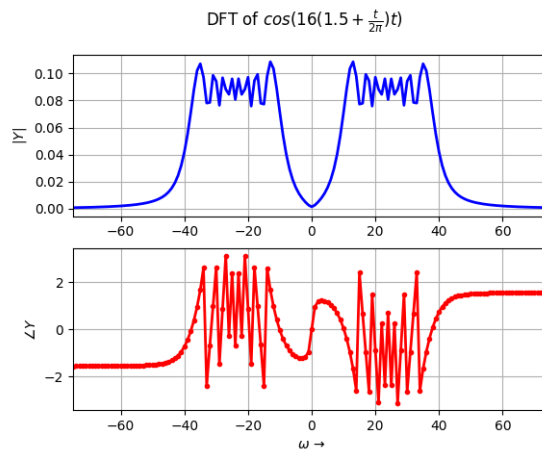
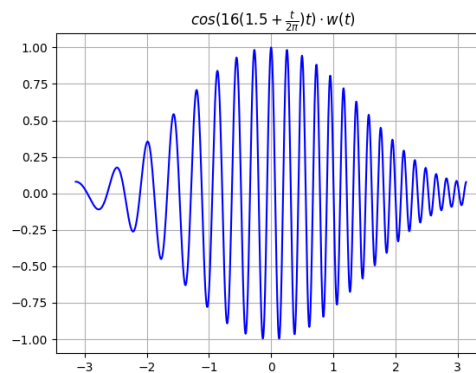
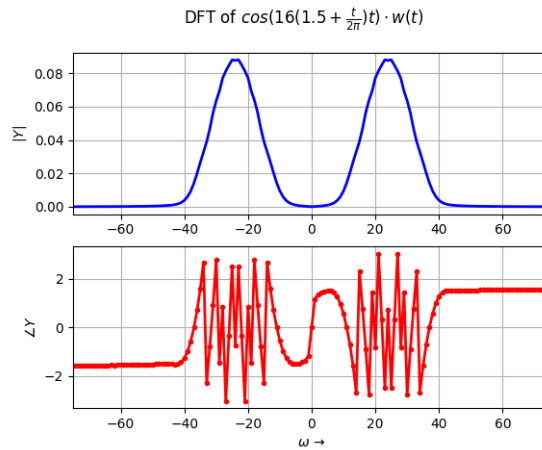
Figure 8:  $\cos(16t(1.5 + \frac{t}{2\pi}))$

We see that the frequency varies from  $16 \text{ rad/sec}$  to  $32 \text{ rad/sec}$  as  $t$  goes from  $-\pi \text{ sec}$  to  $\pi \text{ sec}$ .

On finding the DFT of the above signal, we get:

Applying the Hamming Window to the chirp results in the following:

<sup>1</sup>Source: <https://en.wikipedia.org/wiki/Chirp>

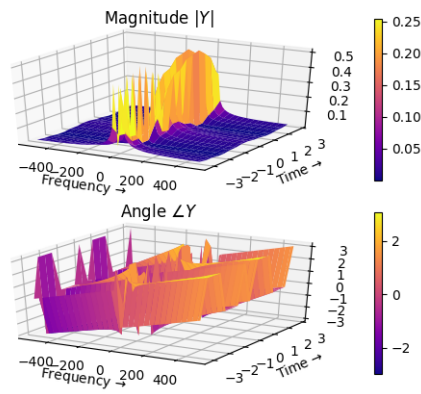
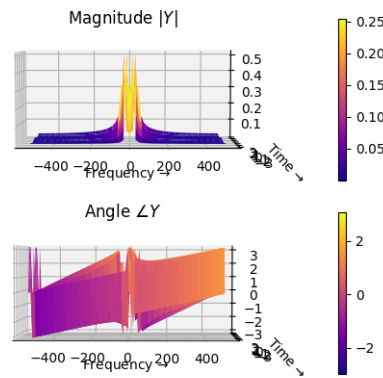
Figure 9: Spectrum of  $\cos(16t(1.5 + \frac{t}{2\pi}))$ Figure 10: Windowed  $\cos(16t(1.5 + \frac{t}{2\pi}))$ Figure 11: Spectrum of windowed  $\cos(16t(1.5 + \frac{t}{2\pi}))$ 

The variations in the frequency have been smoothed out by the window, and also, we can see that the frequencies are more accurately confined to the range 16-32 *rad/sec*.

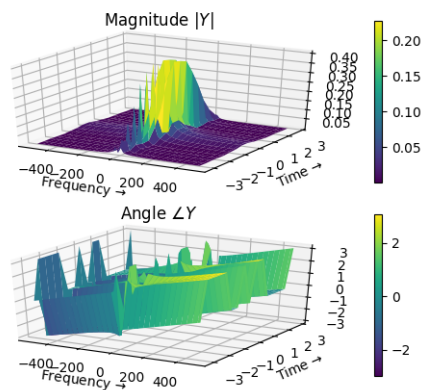
## 6 Time-frequency plot of $\cos(16t(1.5 + \frac{t}{2\pi}))$

We shall split the chirp in the time interval  $[-\pi, \pi]$  into smaller intervals of time, and observe how the frequency of the signal varies with time.

Initially we had a 1024-length vector with the values of the chirp signal. We shall split it into 64-length vectors, take the DFTs of these localized vectors, and plot a time-frequency surface plot to observe the variation of the frequency with time.

Figure 12: Spectrum of windowed  $\cos(16t(1.5 + \frac{t}{2\pi}))$ Figure 13: Spectrum of windowed  $\cos(16t(1.5 + \frac{t}{2\pi}))$ 

Now, we shall do the same, but with a windowed version of the chirp.

Figure 14: Spectrum of windowed  $\cos(16t(1.5 + \frac{t}{2\pi}))$

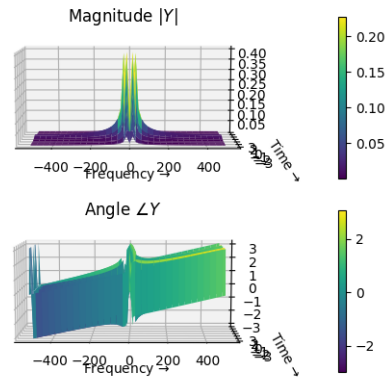


Figure 15: Spectrum of windowed  $\cos(16t(1.5 + \frac{t}{2\pi}))$

We can see that the change in the magnitude is more gradual in the windowed case.

## 7 Conclusion

The DFT was obtained using a  $2\pi$  periodic extension of the signal, and thus the spectrum was found to be erroneous for a non periodic function. The spectrum was rectified by the using a windowing technique, by employing the Hamming window. Given a vector of cosine values in the a time interval, the frequency and phase were estimated from the DFT spectrum, by using the expectation value of the frequency and a parameter tuning for optimum values. The DFT of a chirped signal was analysed and its time-frequency plot showed the gradual variation of peak frequency of the spectrum with time.